the Journal of Strength and Conditioning Research™

Effects of Photobiomodulation on Pain and Return to Play of Injured Athletes: A Systematic Review and Meta-analysis

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Abstract

Morgan, RM, Wheeler, TD, Poolman, MA, Haugen, ENJ, LeMire, SD, and Fitzgerald, JS. Effects of photobiomodulation on pain and return to play of injured athletes: A systematic review and meta-analysis. *J Strength Cond Res* 38(6): e310–e319, 2024—The aims of this systematic review and meta-analysis were to evaluate the effect of photobiomodulation (PBM) on musculoskeletal pain in injured athletes and to determine if the effects of PBM allowed injured athletes to return to play faster. Electronic databases (MEDLINE Complete, CINAHL, and SPORTDiscus, PubMed, Web of Science, and Embase) were systematically searched (up to and including November 7, 2023) for peer-reviewed randomized controlled trials (RCTs) meeting criteria. Six RCTs, representing 205 competitive and recreational athletes with a mean age of 24 years, were included in the analysis. There were 6 intervention groups using standard physical therapy ($n = 1$), placebo PBM ($n = 4$), and aloe gel ($n = 1$) lasting between 10 minutes and 8 weeks in duration. The level of significance set for the study was $p < 0.05$. Overall, the use of PBM indicated a positive effect on pain reduction for PBM vs. control groups, standardized mean differences = 1.03, $SE = 0.22$, 95% confidence intervals = [0.43–1.63], $p = 0.0089$, but the 2 RCTs found evaluating the effect of PBM on time to return to play after injury in athletes do not support a benefit. Allied healthcare professionals may use PBM to reduce pain, thus allowing an athlete to return to their normal biomechanical movement faster; however, limited evidence suggests that PBM does not reduce time to return to play after an injury.

Key Words: physical therapy, modality, performance, recovery, rejuvenate, rehabilitate

Introduction

Photobiomodulation (PBM), also known as low-level laser therapy (LLLT) and light-emitting diode therapy (LEDT), has been suggested to provide several benefits to skeletal muscle tissue (21). Researchers have hypothesized that PBM increases energy metabolism and adenosine triphosphate (ATP) synthesis, stimulates defenses against oxidative stress (37), prevents and repairs skeletal muscle damage, modulates gene expression by activation of transcription factors, and has the potential to increase excitability of muscle fibers (20,21). The primary theory on the positive effects of PBM on skeletal muscle is the stimulation of mitochondria through red or near-infrared (IR) photons (1,21). When the photons are absorbed by cytochrome *c*-oxidase, the following physiologic changes occur in human and animal models: (a) ATP production increases; (b) reduction in inflammation and pain; (c) stimulation of the formation of new muscle fibers; and (d) acceleration of angiogenesis, repair, and regeneration of tissues (1,21,37,38).

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Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (http://journals.lww.com/nsca-jscr).

Journal of Strength and Conditioning Research 38(6)/e310–e319

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Photobiomodulation parameters, such as wavelength and dosing used, seem to modify tissue-specific responses to the therapy. A small dose may lead to an important cellular response, but high doses may inhibit cell proliferation or even cause apoptosis (1). Studies have suggested that wavelength parameters between 600 and 1,070 nm (red/near-IR) have had the best noninvasive effects for cell proliferation (1). Studies have observed shorter wavelengths to be absorbed by hemoglobin or melanin, producing cellular effects, whereas longer wavelengths are absorbed by water and provide a warm sensation and pain relief (1). Many studies have suggested that positive results on the musculoskeletal system were found within a wavelength range of 655–950 nm and an energy dose range from 20 to 60 J for small muscle groups and 60–300 J for large muscle groups and maximal power output of 200 mW per diode (1,56).

Photobiomodulation has been used for treatment of inflammatory diseases (26), immune and nervous system conditions (15,62), traumatic injuries (26), musculoskeletal disorders, stimulation and healing of skeletal muscles, tendons, ligaments, and joints in clinical settings (1,16,55). The results of these studies demonstrated a consistent reduction of inflammation (15,16,26,55,62). Musculoskeletal pain, such as a tendinopathy, is a common condition that can affect a person's physical function and quality of life (54). A meta-analysis by Tripodi et al. (54) found that PBM with exercise reduced pain and increased

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function when compared with sham PBM with exercise. There have been 2 meta-analyses assessing the effectiveness of PBM on the musculoskeletal system of healthy adults, both reporting positive effects on exercise performance when applied before exercises (35,56). To our knowledge, no systematic review and meta-analysis has been conducted on the effects of PBM on musculoskeletal pain and return to play after injury in athletes. Population-specific effect size estimates are needed to guide athletic trainers, physical therapists, physicians, and sports scientists as estimates derived from the general population may not be applicable to highly motivated athletes. Additionally, effect size estimates may also be applicable to fitness professionals who may be looking to add this equipment to their training facility.

The importance of reducing pain and shortening return to play after injury is multidimensional. Although time to return to play depends on the severity of the injury, on average, an injury causes an athlete to miss up to 4 weeks of play (18). The average amount of money spent on sport rehabilitation in college is about $446 million to $1.5 billion USD per year (33). By allowing the athlete to return to play faster by using PBM, it may reduce the cost needed for rehabilitation and medication for pain relief in the future. A systematic review by Bank et al. (8) found that pain negatively affected motor control and movement kinetics to protect the subjects from further pain or injury. Many factors can contribute to altered motor control such as structural damage, emotional responses, and cognitive responses to chronic pain leading to fear-avoidance (58). It seems that both acute and chronic pain may lead to compensatory movement patterns if an athlete in pain returns to a skill or sport performance-based activity, potentially reducing return-to-play readiness and increasing risk for further injury (48). Moreover, an athlete's psychological readiness to return to play is based on the athlete's perspective of the pain, how they behaviorally respond to the pain, and the associated emotions because of the pain (61). Many studies have suggested that fear of re-injury, which is augmented by chronic pain (58), reduces the likelihood that an athlete will return to play (4,5,7,23,36). The aim of this systematic review and meta-analysis was to evaluate the effects of PBM on musculoskeletal pain and return to play after injury in recreational and competitive athletes.

Methods

Experimental Approach to the Problem

This review was guided by the Preferred Reporting Items for Systematic reviews and Meta-Analyses statement (44). Studies were eligible for inclusion if they reported effects of PBM on musculoskeletal pain or return to sport in injured athletes (i.e., competitive and recreational) between the ages of 18 and 44 years and used a randomized controlled trial (RCT) design. Studies needed to be available in the English language and report pre-intervention and postintervention musculoskeletal pain (or changes in pain) or time needed to return to play after injury. Only peer-reviewed journal articles were considered, with all other publication types (e.g., abstracts, reviews, theses, books, and letters) excluded.

Studies were identified by searching electronic databases, reference lists, and topical systematic reviews. A systematic search of the literature was performed on March 29, 2023 in MEDLINE Complete, Cumulative Nursing and Allied Health Literature (CINAHL), SPORTDiscus, PubMed, Web of Science, and Embase. Secondary searches were performed on November 7, 2023 with date restrictions from the March 1, 2023 to November 7, 2023. All sources were limited to the English language. An academic librarian experienced in systematic reviews was consulted during the search strategy development. The librarian helped formulate a search string with Boolean phrases to assist researchers to find relevant articles. This study was not evaluated by the University of North Dakota (UND) Institutional Review Board (IRB) because it only uses descriptive data previously published in RCTs that have undergone institutional review. The descriptive data were obtained from the published articles and no raw data were reviewed or used in this study. Thus, the systematic review does not require institutional review as per UND IRB guidelines.

Procedures

Search terms that were included in a group were combined by the Boolean OR and entered before being combined by the Boolean AND. Selection of search terms was informed by the PICO (population, intervention, comparison, outcome) method. The first group of search terms identified PBM (photobiomodulation* OR PBMT OR phototherap* OR "low-level laser therap*" OR LLLT OR "light-emitting diode therap*" OR LEDT OR "photobiomodulation therap*") and the second group identified the population (athlete* OR sport* OR player*). Supplement 1 shows the full search strategies for each database (see Text, Supplemental Digital Content 1, http://links.lww.com/JSCR/A467).

One researcher executed the electronic database search and imported all results into Zotero. First-level screening involved 2 researchers who independently screened all studies by title and abstract. Second-level screening involved 2 researchers who independently screened all studies that were kept after first-level screening at the full-text level. Each level of screening required both researchers to reach consensus for final inclusion. If the researchers did not come to consensus, then a third researcher resolved any discrepancies. Consensus was achieved among all 3 researchers through discussion when discrepancies existed. Figure 1 demonstrates the process used to retain a total of 6 articles. Two researchers who published on the effects of PBM on pain in injured athletes were contacted to locate additional studies and referred studies were evaluated against inclusion criteria. One researcher reviewed the references of relevant articles and systematic reviews in addition to searching Google Scholar's top 200 results with the identical Boolean phrases that were used in the electronic databases above.

Study methodological quality and completeness was appraised using the Physiotherapy Evidence Database (PEDro) tool for RCTs. Scores range from 0 to 10 using the PEDro tool. Low, moderate, and high quality was inferred if scores were 0–3, 4–5, and 6–10, respectively (40). This tool has good reliability (40). The quality assessment was independently carried out by 2 reviewers. Interrater agreement was assessed using Cohen's Kappa (41).

Statistical Analyses

Meta-analyses were performed in Meta-Essentials (Erasmus Research Institute of Management, Erasmus University Rotterdam, The Netherlands) (51). The primary analysis was conducted to determine an overall effect of musculoskeletal pain reduction when using PBM. This systematic review only had 2 studies that assessed return to play, so a meta-analysis was not performed for this outcome. The magnitude and direction of effect size estimates from synthesizing data from less than 3 studies does not accurately forecast estimates when more studies are available and should not be used for decision making (63). The level of significance set for the study was $p < 0.05$. A random effects model was used for each

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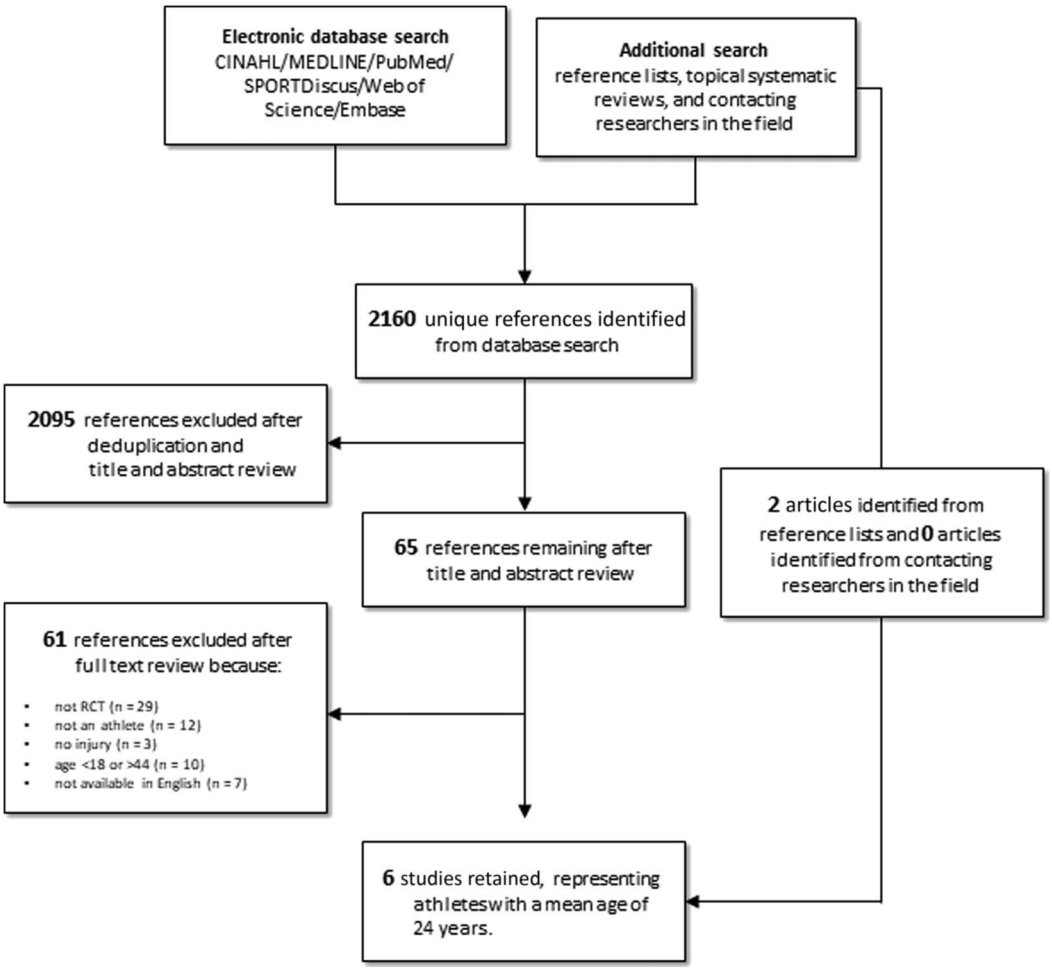

Figure 1. Prisma diagram showing flow of studies through the systematic review.

study to calculate the standardized mean differences (SMD) and 95% confidence intervals (CIs) for each intervention group weighted by the inverse of the pooled variance. Calculating the SMD was important to be able to measure the overall effect on musculoskeletal pain, even if different scales were used in the studies that were included in this systematic review. The weighted mean difference method was used to calculate experimental effects relative to the control group. Mean differences that were positive indicated a favorable experimental effect, and mean differences that were negative indicated an unfavorable experimental effect. Standardized mean differences were interpreted using Cohen's (13) thresholds of 0.2, 0.5, and 0.8 for small, medium, and large, respectively. Statistical heterogeneity was quantified using the I^2 statistic, with values of 0–30% (might not be important), 30–60% (moderate), 50–90% (substantial), and 75–100% (considerable) (29). In this study, I^2 was equal to 46.87%. A funnel plot asymmetry analysis was not performed because we had fewer than 10 comparison groups, which compromises the power of the test to detect real asymmetry (28).

Results

Study Selection

There were a total of 1,090 articles that were identified initially after deduplication. Sixty-five studies were retained after the first level of screening and 4 articles met criteria after the second level of screening. Two researchers who published on the effects of PBM on pain in injured athletes were contacted to locate additional studies and referred studies were evaluated against inclusion criteria. After searching the reference list of relevant articles and Google Scholar, 2 additional articles were found making the total number of articles included to be 6 (3,42,50,52,53,57). Figure 1 demonstrates the process used to retain included studies.

Study Characteristics

All RCTs were published in English between 2004 and 2022, collectively representing 205 competitive and recreational athletes with a mean age of 24 years. Overall, there were 6 intervention groups using standard physical therapy (n = 1), placebo PBM (n = 4), and aloe gel (n = 1) lasting between 10 minutes and 8 weeks in duration. Study characteristics are presented in Table 1.

Methodological Quality

The 2 reviewers assessing methodological quality with the PEDro scale were in perfect agreement (Cohen's Kappa = 1). Included studies were high quality with scores ranging from 6/10 to 10/10.

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Table 1

Description of parameters and outcomes for included studies.*

Study	Athletes	Injury	E (n)	C (n)	Outcome measures	Comparison	Type of laser used	Modality settings	Intervention duration	Intervention description	Outcome	PEDro score
Ammendolia et al. (3)	Volleyball players	Tendinitis	16	19	Pain (VAS), resumption of gym training, and time to return to play	Topical use of an aloe gel (C) vs. LLLT (E)	Not specified	Group C: 3 ml gel based on aloe barbadensis extract (Alevix gel—Vita Research s.r.l.—Italy), 3 times a day. Group E: wavelength = 904 nm; power = 25 mW·cm^{-2} in pulsed mode for a duration of 3 min.	10 d	Group C: 3 daily topical applications of an aloe gel Group E: 1 daily session of LLLT	In all athletes of both groups, we observed a progressive reduction of the pain, but more significant in patients with an Achilles tendinitis ($p < 0.004$), treated with gel. There was a statistically significant difference ($p < 0.0001$) between the groups about the time elapsed since the start of the treatment and the restart of the training in gym and the return to play volley, especially in athletes treated with gel. Gel out performed LLLT; however, both modalities reduced pain and reduced the number of days needed to return to play. LLLT: return to play was 11 d on average/Gel: return to play was 8 d on average.	6
Medeiros et al. (42)	Male ameteur athletes	Hamstring	11	11	Primary: time to return to sport; secondary: no. of rehab sessions, hamstring flexibility, strength, re-injury rate	LLLT vs. placebo	Chattanooga intelect advanced with a cluster probe consisting of 5 infrared diodes	Number of sites: 3; treatment time (per site): 60 s; number of diodes (cluster probe): 5; wavelength: 850 nm; frequency: continuous; probe size: 7.5 cm^2; spot size (per diode): 0.029 cm^2; peak power (per diode): 100 mW; power density (per diode): 3.44 mW·cm^{-1}; energy	The treatment period and, consequently, the number of rehabilitation sessions, varied among the subjects because their discharge was defined by specific criteria to return to sport.	Hamstring muscles were treated with LLLT or placebo immediately after each rehabilitation session. The rehabilitation sessions were 3×/wk, with at least 48 h between sessions and comprised 3 phases to enhance hamstring strength, trunk stabilization, and	The 2 groups increased flexibility and strength similarly throughout the rehabilitation program. Time-to-return to sport was the same for athletes treated with LLLT (23 ± 9 d) and placebo (24 ± 13 d). There were no re-injuries within 6 mo after return to sport.	10

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Table 1

Description of parameters and outcomes for included studies.* *(Continued)*

Study	Athletes	Injury	E (n)	C (n)	Outcome measures	Comparison	Type of laser used	Modality settings	Intervention duration	Intervention description	Outcome	PEDro score
								density (per diode): 206.9 J·cm^{-2}; energy (per diode): 6 J; energy (per site): 30 J; energy (per leg): 90 J; placebo group had the same parameters; however, the machine was turned off.		movement agility. Exercises were specific for each rehabilitation phase, whereas exercise workloads were individualized for each subject. The uninjured limb performed the same exercises to avoid detraining effects. To progress to the next phase, subjects should meet pre-established criteria.	Subjects from both groups attended a similar number of physiotherapy sessions: LLLT group, 9.18 ± 3.31; placebo group, 9.36 ± 4.78. Finally, time-to-return to sport was similar between groups: LLLT group, 23.09 ± 9.08 d; placebo group, 23.82 ± 12.62 d. At 6 mo follow-up, all subjects declared to have returned to their sports at the pre-injury level and none of them had sustained re-injuries.	
Stergioulas et al. (50)	Recreational athlete	Achilles tendinopathy	20	20	Pain with activity using 100-mm VAS; secondary: morning stiffness severity, measured on a 100-mm VAS; crepitation, measured on a 100-mm VAS; tenderness on a 40-mm VAS; and range of active ankle dorsiflexion in degrees, measured by goniometry	Ecc + LLLT or Ecc + placebo LLLT	Emitting infrared laser light from a laser probe with Ga-Al-As diode (Biotherapy 2000; Omega Universal Technologies, London, United Kingdom)	Power density: 60 mW·cm^{-2}; total dose: 5.4 J per session; wavelength: 820 nm; optical output: 30 mW; irradiation area/spot size: 0.5 cm^2; power density on skin: 60 mW·cm^{-2}; number of irradiation points: 6; energy per point in each session: 0.9 J; total energy per session: 5.4 J; total energy delivered in 12 sessions: 64.8 J.	8 wk, 12 sessions	Ecc regimen was performed with supervision 4×/wk for 8 wk. The regimen was to complete 12 sets of 12 repetitions with 1-min rest between sets, 4 d/wk for 8 wk. Athletes started with 1 × 5 on day 1 and progressed to 8 × 12 repetitions, aiming to complete 12 × 12 repetitions by the fourth week of treatment. All patients received 2 treatments per week of LLLT in each of the first 4 wk and 1 treatment per week in each of the second 4 wk.	Pain intensity during physical activity on the 100-mm visual analog scale was significantly lower in the LLLT group than in the placebo LLLT group, with 53.6 vs. 71.5 mm ($P = 0.0003$) at 4 wk, 37.3 vs. 62.8 mm ($P = 0.0002$) at 8 wk, and 33.0 vs. 53.0 mm ($P = 0.007$) at 12 wk after randomization. The secondary outcomes were significantly better ($P < 0.05$) in the laser group than in the placebo groups at 4, 8, and 12 wk after randomization. All 40 patients completed all 12 sessions of LLLT or placebo LLLT and had	7

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Table 1

Description of parameters and outcomes for included studies.* *(Continued)*

Study	Athletes	Injury	E (n)	C (n)	Outcome measures	Comparison	Type of laser used	Modality settings	Intervention duration	Intervention description	Outcome	PEDro score
											a compliance of 85–100% to the Ecc regimen.	
Takenori et al. (52)	College athletes	E: 9 ankle sprain, 1 navicular fx, 1 plantar fasciitis, 1 patella tendinitis, 1 spondylolysis, 1 shoulder arthroscopic surgery, 1 TFCC injury, 1 proximal thumb avulsion fx; C: 5 ankle sprain, 2 meniscal injuries, 2 elbow medial collateral ligament sprain, 2 achilles tendinitis, 1 LBP, 1 lumbar facet arthritis, 1 infraspinatus muscle injury, 1 deltoid muscle injury, 1 shoulder periarthritis	16	16	Pain using the modified numerical rating scale	LLLT vs. placebo	Laser group: Softlasery (JQ-W1, Minato Medical Science Co., Ltd., Japan [GaAlAs laser]) used for this study was contact-type laser therapy equipment Placebo group: detuned laser	Laser group: wavelength: 810 nm; frequency: continuous output; optical output: 180 mW; spot diameter: 0.0007 cm, 0.0005 cm; spot size: 0.0035 cm^2; power density: 51.4 $W{\cdot}cm^{-2}$; energy: 5.4 J at each spot; energy density: 1,542.85 $J{\cdot}cm^{-2}$ at each spot; treatment time: 30 s × 20 times (total 10 min); irradiation site: 1 cm^2.	Laser group: 10 min; placebo group: 10 min	Application mode: probe held stationary in skin contact with a 90° angle and slight pressure; placebo group: 0 mW, 30-s irradiation.	Low-level laser therapy was effective in 75% of the laser group, whereas it was not effective in the placebo group, indicating a significant difference in favor of the laser group ($p < 0.001$). Pain relief rate was significantly higher in the laser group than in the placebo group (36.94 vs. 8.20%, respectively, $p < 0.001$), with the difference in pain relief rate being 28.74%.	8
Taleb et al. (53)	Handball athletes	Groin strain	20	20	HAGOS, pressure algometry (PPT), CK, and lactate dehydrogenase levels	Diode laser, TBT, and medical treatment vs. Sham laser, TBT, and medical treatment	Chattanooga gallium aluminum arsenide diode laser	905 nm wavelength.	3×/wk for 12 sessions day after day over 4 wk	E: Laser, Ambezim-G 5 mg, and TBT; C: sham laser with same medical and TBT treatment.	The pressure algometer at point 1 and point 2 showed significant improvements in both groups after treatment, where the main effect of time was ($p < 0.0001$) for both but, the experimental group showed higher significant improvement than the control group posttreatment as the group main effect was ($p < 0.005$ and $p < 0.003$), respectively, and time × group interaction effect was ($p < 0.0001$) for both.	8

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Table 1

Description of parameters and outcomes for included studies.* *(Continued)*

Study	Athletes	Injury	E (n)	C (n)	Outcome measures	Comparison	Type of laser used	Modality settings	Intervention duration	Intervention description	Outcome	PEDro score
Verma et al. (57)	Track and field athletes	Proximal hamstring tendinoptahy	18	18	IPT of hamstring (isokinetic dynamometer), pain (NPRS)	HPLT vs. coventional PT	LASER equipment (LiteCure)	Average output power: 5 watts; dosage: 50 J·cm^{-2}; wavelength: 980/810 nm; total treatment area: 6 × 6 cm = 36 cm^2; total energy: 50 × 36 = 1800 joules. The total treatment time was to be 6 min on continuous mode. HPLT 3 d·wk^{-1}.	3 wk	The experimental group included the application of HPLT for 3 wk. The conventional group included treatment with a conventional PT program including ultrasound therapy, moist heat pack, and home exercises for a total of 3 wk.	Pain score decreased, and IPT increased significantly ($p < 0$: 05) after application of HPLT, by 61.26 and 13.18%, respectively. In the conventional group, a significant difference ($p < 0$:05) was observed in pain scores only, which decreased by 41.14%. When HPLT was compared with conventional PT, a significant difference was found in pain scores only. HPLT for 3 wk was found to be effective in improving pain in athletes with proximal hamstring tendinopathy. However, no significant difference was found between HPLT and conventional PT (US, moist heat, and home exercises) in improving the IPT of the hamstring muscle.	8

*C = control group; CK = creatine kinase; E = experimental group; Ecc = Eccentric strengthening; fx = fracture; HAGOS = Copenhagen Hip and Groin Outcome Score; HPLT = high-power laser therapy; IPT = isokinetic peak torque; LBP = low back pain; LEDs = light-emitting diodes; LLLT = low-level laser therapy; LOS = limits of stability; NPRS = numeric pain rating scale; PEDro = Physiotherapy Evidence Database; PEDro scores range from 0 to 10 (0–3 = low quality, 4–5 = moderate quality, and 6–10 = high quality); PT = physical therapy; PPT = pain pressure threshold; PST = postural stability testing; TBT = trigger band technique; TFCC = triangular fibrocartilage complex; VAS = visual analogy scale; (+) = Positive.

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Synthesis of Results

Musculoskeletal pain rating scale and time to return-to-play data were collected. Six studies evaluated pain by the visual analogue scale (3,50), modified numerical rating scale (52), pain pressure threshold (53), and numeric pain rating scale (57). The findings indicated a positive effect on pain reduction for PBM vs. control groups, SMD = 1.03, SE = 0.22, 95% CI = [0.43–1.63], p = 0.0089 (Figure 2).

The 2 studies that evaluated time to return to play in number of days will be summarized in narrative because meta-analysis is not appropriate for 2 comparison groups (63). One study suggested that an aloe gel group returned the athlete to play quicker than the PBM group (3). The other study suggested that there was no significant difference between the PBM group and the placebo group in the days it took for the athlete to return to play (42).

Discussion

Achieving pain reduction in athletes is important because it allows the athlete to progress their rehabilitation, most likely allowing them to return to play faster (19). The findings from our meta-analysis of 5 RCTs indicate that PBM has a large, positive effect on musculoskeletal pain in athletes rehabilitating from injury. Managing pain more effectively may reduce its negative effect on motor control, movement kinetics, force production (8), and fear avoidance (58) during rehabilitation, speeding return to play (4,5,7,23,36). Despite this large effect on pain, the 2 RCTs found evaluating the effect of PBM on time to return to play after injury in athletes did not support a benefit.

In comparison with this systematic review, other systematic reviews of RCTs have also demonstrated positive effects of PBM in both shorter (2 weeks) (35) and longer duration studies (2–6 months) (31,47). According to Tripodi et al. (54), there was a positive, substantial effect for reduction in pain in tendinopathy groups that completed PBM with exercise when compared with groups that completed sham PBM with exercise. Penberthy and Vorwaller (45) found a large effect for reduction in pain and improved functional outcome measures when using PBM at 1,064 nm. A systematic review by Vanin et al. (56) found a positive, moderate effect of PBM on exercise performance (i.e.,—strength, power, and endurance) and muscular fatigue in healthy individuals during exercise when compared with a placebo group. Most of the positive results were found at a wavelength between 655 and 950 nm (56). A meta-analysis by Borsa et al. (10) assessed skeletal muscle contractile function and post-exercise recovery. The authors found that using PBM before exercise demonstrated ergogenic effects and prophylaxis effects when compared with a placebo group (10). The exercise performance (i.e., repetitions, time to fatigue, work, strength loss, strength recovery, and EMG) effect size was moderate and the prophylaxis (i.e., biomarkers for muscle damage such as CK levels, lactate dehydrogenase levels, CRP, and blood lactate) effect size was large in favor of PBM (10). Finally, a meta-analysis by Leal-Junior et al. (35) found a large, positive effect on exercise performance (number of repetitions) and muscle recovery (time to exhaustion). These meta-analyses reporting moderate-to-large positive effects of PBM before exercise evidence an effect of PBM on musculoskeletal function (10,35,56). Consistent with these findings, our study found large effects of PBM on reducing musculoskeletal pain in athletes after injury.

The majority of studies (4 of 5) included in our meta-analysis evaluated the long-term effects of PBM on pain, although PBM can have short-, medium-, and long-term effects on pain (14). In the 1 study included in our analysis assessing the short-term effects of PBM, Takenori et al. (52) reported acute reductions in pain 10 minutes after PBM treatment (wavelength = 810 nm) compared with placebo. The acute effect of PBM on pain is likely from directly inhibiting action potentials of the noxious stimuli of the epidermal neural network, which then affects the subcutaneous tissues, sympathetic ganglia, and neuromuscular junctions, reducing pain and muscle spasms within minutes (9,14). This inhibition decreases the release of pro-inflammatory neuropeptides (i.e., substance P and CGRP), therefore reducing pain (14). No studies were found meeting our criteria and assessing medium-term effects, which seem to be because of a decrease in local edema and reduction of inflammation within hours to days as inflammatory cells are reduced in joints (2,6,11,24,25,39,46). Our meta-analysis best reflects the long-term effects of PBM on pain that are thought to occur within 1 to 2 weeks and can last for months to years as a result of improved tissue healing (14). Three of 4 of those studies found PBM (wavelength range = 810–980 nm) to reduce pain in athletes when compared with control groups (all 4 studies). Our findings seem to mirror that in clinical patients. Langella et al. (34) and Herpich et al. (27) used wavelengths between 640 and 905 nm and found a decrease in pain by the visual analog scale and a decrease in TNF-α and IL-8 serum levels (34). A decrease in TNF-α and IL-8 serum levels also decreases the pro-inflammatory response, therefore causing pain to reduce (34). The other 2 studies found a reduction in pain in patients with knee osteoarthritis (49) and a reduction in pain with temporal mandibular dysfunction (27) by reduction in inflammatory cytokines as well. There have been numerous animal studies that support the decrease in pain by reducing inflammatory cytokines at a wavelength of 810–980 nm (12,17,30).

The mechanism of PBM effect and its ability to modulate pain does not seem to be strong enough to achieve reduction in return to play, although our evidence is tentative. One study compared return to play between a topical aloe gel group and a PBM group

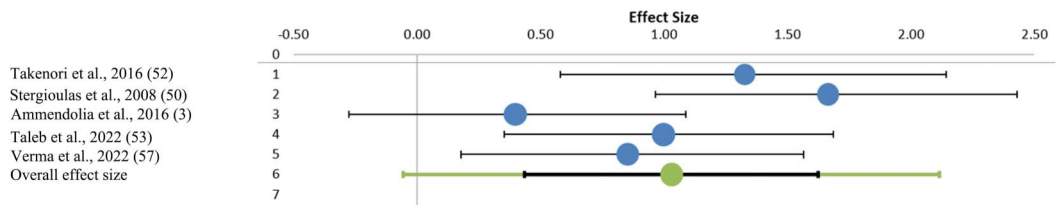

Figure 2. Forest plot of the overall effects of photobiomodulation (PBM) on pain reduction in injured athletes. The blue circles represent the standardized mean differences (SMD) of each individual study and the green circle represents the overall effects of PBM on pain reduction in injured athletes. Positive SMDs indicate favorable experimental effects and negative SMDs indicate unfavorable experimental effects.

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with Achilles tendinitis (3). The topical aloe gel group returned athletes to play in about 8 days on average where the PBM group returned athletes to play in 11 days (3). Typically, return to play after being diagnosed with an Achilles tendinopathy is a minimum of 12 weeks (32,43). The aloe gel used in this study was not solely aloe vera. It consisted of other micronutrients, such as extract of *Arnica montana*, extract of *Harpagophytum procumbens*, and tocopherol (3). *Arnica montana*, *H. procumbens*, and tocopherol all have anti-inflammatory properties (22,59,60). Thus, it is difficult to evaluate the effectiveness of PBM in the context of this study. The second study compared PBM against a placebo control group with hamstring injuries (42). This study found that there was no difference in the time to return to play between the groups with each group taking about 24 days on average to return to play (42). Both groups completed the same hamstring rehabilitation program that focused on hamstring strength, trunk stabilization, and movement agility (42). It should be noted that severity of hamstring injury data were not collected in this study and could be different between groups. Also, this study was powered to detect large effects and the effect of PBM on return to play may be small.

This systematic review has limitations. Our meta-analysis of musculoskeletal pain only had 5 studies that reduces the confidence in the precision of our effect size estimate. Return-to-play data were not meta-analyzed because of limited RCTs meeting our inclusion criteria. With only 2 studies with differing approaches to treatment and control groups, it is difficult to evaluate the potential of PBM to reduce return to play after injury. Finally, the review protocol for this study was a part of a broader performance-based project and was not prospectively registered with the International Prospective Register of Systematic Reviews.

In conclusion, PBM has a large, positive effect on reducing pain in athletes. Limited evidence suggests that PBM has no effect on return to play. More research is required to determine if PBM can return an athlete to play faster after an injury when compared with no treatment.

Practical Applications

Our meta-analysis indicates that PBM can reduce pain in athletes after injury. Those responsible for acquiring equipment should select devices that emit wavelengths between 810 and 980 nm because these were used in the included studies. Commercially available PBM handheld devices range in price from $150 to $30,0000 USD, with whole-body PBM equipment (red-light bed) costing roughly $125,000 USD. Athletic trainers or physical therapists administering the treatment with the aim of reducing musculoskeletal pain should apply PBM for at least 3 minutes, but no longer than 10 minutes, 3–4×/week for at least 3–8 weeks. Achieving meaningful reductions in pain should theoretically allow the injured athlete to progress with their supervised training program quicker, minimizing the negative effects of chronic pain on motor control, movement kinetics, force production, adaptation, task performance, and fear avoidance. More research is needed to see if PBM can reduce the time an athlete spends away from their sport after injury. Researchers designing PBM trials in injured athletes should recruit larger cohorts, assess injury severity, use differing PBM dosing strategies, use standard performance assessments and report return-to-play assessment criteria. Examining psychological readiness for return to play alongside physical readiness also has clinical utility because it is associated with not returning to sport (5) or returning to sport at lower levels of performance (4,23). Future research should also evaluate the effectiveness of whole-body PBM equipment (red-light bed) on pain as they are commercially available, easy to operate, and do not require trained personnel to operate, reducing time burden on practitioners and increasing access to the beneficial effect of PBM (e.g., sport teams and fitness facilities). Sports scientists should continue to monitor the literature for studies addressing the above deficiencies in the current limited evidence of the effect of PBM on return to play.

Acknowledgments

The authors have no conflicts of interest to disclose. The results of the present study do not constitute endorsement of the product by the authors or the NSCA. There was no funding provided to complete this article.

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